

Functional Renormalization Group Notes

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Abstract

These notes collect two related pieces of functional-renormalization-group machinery: the standard fermionic fRG setup with a cutoff-dependent quadratic kernel, single-scale propagator, and Katanin substitution; and the spin fRG construction for an anisotropic Heisenberg model, including deformation of longitudinal/transverse couplings, hybrid functionals, spontaneous symmetry breaking, and the vertex hierarchy. This is a working note. Signs, regulator conventions, factors of βN , and the exact meaning of the Katanin replacement should be checked.

1 Fermionic fRG: minimal setup

Start from a Grassmann functional integral

$$\mathcal{Z} = \int \mathcal{D}(\bar{\psi}, \psi) e^{-S(\bar{\psi}, \psi)}. \quad (1)$$

Separate the quadratic and interacting parts as

$$S(\bar{\psi}, \psi) = (\bar{\psi}, Q\psi) - \tilde{V}(\bar{\psi}, \psi), \quad (\bar{\psi}, Q\psi) = \sum_j \bar{\psi}_j Q_j \psi_j. \quad (2)$$

For a translationally invariant fermion problem the composite index is typically $j = (i\omega_n, \mathbf{k}, s)$ and

$$Q_j = i\omega_n - \xi_{\mathbf{k}}, \quad \xi_{\mathbf{k}} = \varepsilon_{\mathbf{k}} - \mu. \quad (3)$$

The inverse of the quadratic kernel is the bare propagator, $G_0 = Q^{-1}$, up to sign and normalization conventions.

2 Cutoff and single-scale propagator

Introduce a cutoff-dependent quadratic kernel

$$Q^\Lambda = Q/\chi(\Lambda), \quad \chi(\Lambda) \approx \Theta(|\xi| - \Lambda). \quad (4)$$

The flow starts near a bandwidth scale $\Lambda_i \sim W$ and ends at $\Lambda_f = 0$, where $\chi(0) = 1$. The full propagator is

$$G^\Lambda = (Q^\Lambda - \Sigma^\Lambda)^{-1}. \quad (5)$$

The single-scale propagator is the derivative of G^Λ at fixed self-energy:

$$S^\Lambda = -G^\Lambda \dot{Q}^\Lambda G^\Lambda, \quad \dot{Q}^\Lambda = \partial_\Lambda Q^\Lambda. \quad (6)$$

For a sharp-shell cutoff, $\dot{\chi}$ produces a delta function on the running shell $|\xi_{\mathbf{k}}| = \Lambda$.

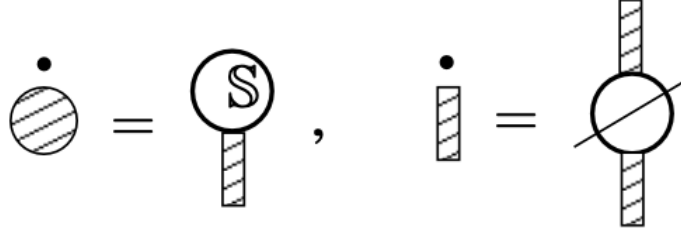


Figure 1: Schematic self-energy and two-particle vertex flow. The vertex flow uses the Katanin modification.

3 Katanin replacement

At the usual one-loop truncation level, the self-energy and two-particle vertex obey closed flow equations after higher vertices are either neglected or approximated. Diagrammatically, the self-energy flow contains one single-scale line and one two-particle vertex, while the two-particle vertex flow contains particle-particle and particle-hole loop corrections.

The Katanin replacement promotes the single-scale structure in the two-particle flow according to

$$S^\Lambda G^\Lambda + G^\Lambda S^\Lambda \longrightarrow \partial_\Lambda (G^\Lambda G^\Lambda). \quad (7)$$

Equivalently, this accounts for the self-energy feedback term contained in $\partial_\Lambda G^\Lambda$ inside the corresponding two-particle loop.

Convention warning. Do not globally replace every S^Λ by $\partial_\Lambda G^\Lambda$ without checking the truncation. The Katanin substitution is a controlled modification of the two-particle flow.

4 Summation trick

Momentum sums can often be reduced to energy integrals,

$$\sum_{\mathbf{k}} \longrightarrow \int_{-W}^W d\varepsilon \rho(\varepsilon). \quad (8)$$

If the scale dependence enters only through χ and Σ , then for an integrand $\mathcal{I}(\xi, \chi, \Sigma)$,

$$\int_{-W}^W d\varepsilon \rho(\varepsilon) \partial_\Lambda \mathcal{I} = \int_{-W}^W d\varepsilon \rho(\varepsilon) \left(\dot{\chi} \frac{\partial \mathcal{I}}{\partial \chi} + \dot{\Sigma} \frac{\partial \mathcal{I}}{\partial \Sigma} \right). \quad (9)$$

For a sharp cutoff this becomes

$$\int d\varepsilon \rho(\varepsilon) \left[\mathcal{I}|_{\chi=1} \delta(|\varepsilon| - \Lambda) + \dot{\Sigma} \frac{\partial \mathcal{I}}{\partial \Sigma} \right]. \quad (10)$$

5 Spin fRG deformation scheme

For the anisotropic Heisenberg model, write

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{V}, \quad (11)$$

with

$$\mathcal{V} = -\frac{1}{2} \sum_{ij} \left[J_{ij}^{\perp} \left(S_i^+ S_j^- + S_i^- S_j^+ \right) + J_{ij}^z S_i^z S_j^z \right]. \quad (12)$$

The deformation parameter Λ is introduced through

$$J_{\Lambda,ij}^{\perp} = J_{ij}^{\perp} - R_{\Lambda,ij}^{\perp}, \quad J_{\Lambda,ij}^z = J_{ij}^z - R_{\Lambda,ij}^z. \quad (13)$$

The regulator is chosen so that the initial point is solvable and the final point recovers the physical model.

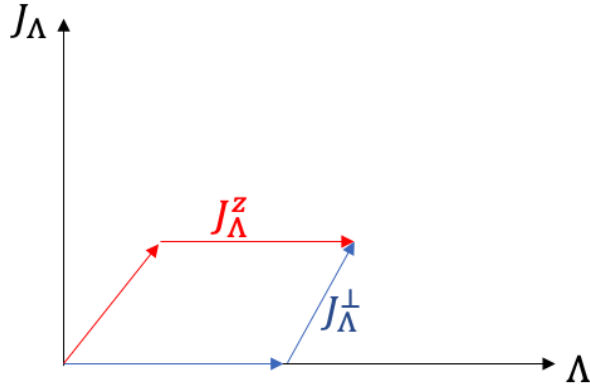


Figure 2: Example deformation: longitudinal fluctuations are switched on first, followed by transverse fluctuations.

6 Generating functional and exact flow

Define the connected generating functional

$$\mathcal{G}_{\Lambda}[\mathbf{h}] = \ln \text{Tr} \left[e^{-\beta \mathcal{H}_0} \mathcal{T} \exp \left\{ \int_0^{\beta} d\tau \left[\sum_i \mathbf{h}_i(\tau) \cdot \mathbf{S}_i(\tau) - \mathcal{V}_{\Lambda}(\tau) \right] \right\} \right]. \quad (14)$$

Connected spin correlators are

$$G_{\Lambda, I_1 \dots I_n}^{\alpha_1 \dots \alpha_n} = \left. \frac{\delta^n \mathcal{G}_{\Lambda}[\mathbf{h}]}{\delta h_{I_1}^{\alpha_1} \dots \delta h_{I_n}^{\alpha_n}} \right|_{h=0}. \quad (15)$$

After a modified Legendre transform, the irreducible functional obeys

$$\partial_{\Lambda} \mathcal{L}_{\Lambda}[\mathbf{M}] = \frac{1}{2} \text{Tr} \left[\left(\mathcal{L}_{\Lambda}''[\mathbf{M}] + \mathcal{R}_{\Lambda} \right)^{-1} \partial_{\Lambda} \mathcal{R}_{\Lambda} \right]. \quad (16)$$

7 Hybrid functional and longitudinal amputation

A Hubbard-Stratonovich decoupling of the longitudinal sector motivates a hybrid functional

$$\mathcal{F}_\Lambda[\mathbf{h}^\perp, s] = \mathcal{G}_\Lambda \left[\mathbf{h}^\perp, h_i^z = \sum_j J_{\Lambda,ij}^z s_j \right] + \frac{1}{2} \sum_{IJ} \mathbb{J}_{\Lambda,IJ}^z s_I s_J. \quad (17)$$

Its derivatives define objects connected in the transverse sector and amputated in the longitudinal sector. For example,

$$F_{\Lambda,I}^z = \sum_J \mathbb{J}_{\Lambda,IJ}^z G_{\Lambda,J}^z, \quad (18)$$

$$F_{\Lambda,IJ}^{zz} = \sum_{KL} \mathbb{J}_{\Lambda,IK}^z \mathbb{J}_{\Lambda, JL}^z G_{\Lambda,KL}^{zz} + \mathbb{J}_{\Lambda,IJ}^z. \quad (19)$$

8 Symmetry-broken phase

In the ordered phase,

$$\langle S_I^\alpha \rangle_\Lambda = M_\Lambda \delta^{\alpha z}. \quad (20)$$

In the hybrid representation the corresponding longitudinal exchange field is

$$\bar{\Phi}_{I,\Lambda}^\alpha = \phi_\Lambda \delta^{\alpha z}, \quad \phi_\Lambda = J_{\Lambda,0}^z M_\Lambda. \quad (21)$$

9 Tree-level two-point identities

In momentum-frequency space, $K = (\mathbf{k}, i\omega)$,

$$F_\Lambda^{+-}(K) = \left[\Gamma_\Lambda^{+-}(K) + R_{\Lambda,k}^\perp \right]^{-1}, \quad (22)$$

$$F_\Lambda^{zz}(K) = \left[\Gamma_\Lambda^{zz}(K) + R_{\Lambda,k}^\phi \right]^{-1}. \quad (23)$$

The transverse connected and partially amputated functions coincide,

$$G_\Lambda(K) = G_\Lambda^{+-}(K) = F_\Lambda^{+-}(K), \quad (24)$$

whereas the longitudinal relation is

$$F_\Lambda(K) = F_\Lambda^{zz}(K) = (J_{\Lambda,k}^z)^2 G_\Lambda^{zz}(K) + J_{\Lambda,k}^z. \quad (25)$$

Thus,

$$\Gamma_\Lambda^{+-}(K) = G_\Lambda^{-1}(K) + J_{\Lambda,k}^\perp - J_k^\perp, \quad (26)$$

$$\Gamma_\Lambda^{zz}(K) = -\frac{G_\Lambda^{zz}(K)}{1 + J_{\Lambda,k}^z G_\Lambda^{zz}(K)} + \frac{1}{J_k^z}. \quad (27)$$

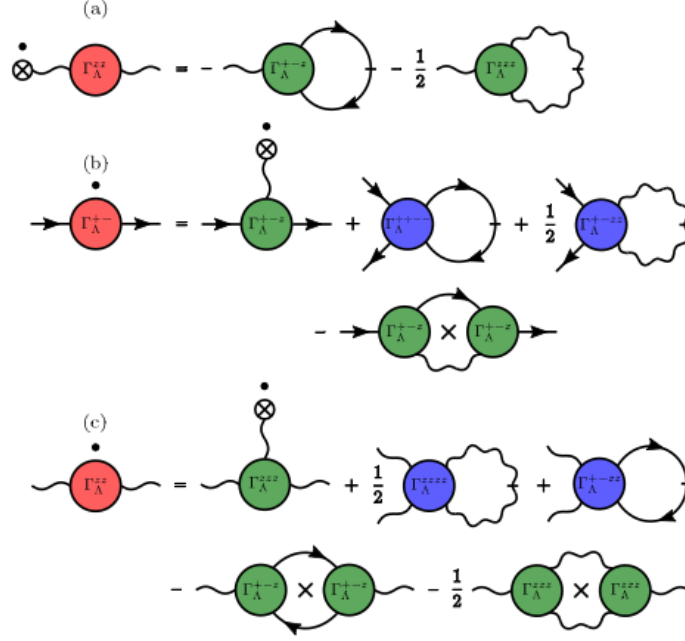


Figure 3: Representative vertex-flow hierarchy diagram from the original notes.

10 Vertex-flow hierarchy

Expanding the exact functional flow produces an infinite hierarchy of coupled integro-differential equations for the vertices $\Gamma_{\Lambda, K_1 \dots K_n}^{\alpha_1 \dots \alpha_n}$.

The free-energy flow is schematically

$$\partial_{\Lambda} f_{\Lambda} = \int_K G_{\Lambda}(K) \partial_{\Lambda} R_{\Lambda, k}^{\perp} + \frac{1}{2} \int_K [F_{\Lambda}(K) - J_{\Lambda, k}^z] \partial_{\Lambda} R_{\Lambda, k}^{\phi} - \frac{1}{2} \phi_{\Lambda}^2 \partial_{\Lambda} R_{\Lambda, 0}^{\phi}. \quad (28)$$

The exchange-field flow is fixed by removing the linear fluctuation term:

$$\Gamma_{\Lambda}^{zz}(0) \partial_{\Lambda} \phi_{\Lambda} = -\partial_{\Lambda} \left(R_{\Lambda, 0}^{\phi} \phi_{\Lambda} \right) - \int_K \dot{G}_{\Lambda}(K) \Gamma_{\Lambda}^{+-z}(K, -K, 0) - \frac{1}{2} \int_K \dot{F}_{\Lambda}(K) \Gamma_{\Lambda}^{zzz}(K, -K, 0). \quad (29)$$

The transverse and longitudinal single-scale propagators are

$$\dot{G}_{\Lambda}(K) = -G_{\Lambda}^2(K) \partial_{\Lambda} R_{\Lambda, k}^{\perp}, \quad \dot{F}_{\Lambda}(K) = -F_{\Lambda}^2(K) \partial_{\Lambda} R_{\Lambda, k}^{\phi}. \quad (30)$$

11 Infinite-range / Katanin benchmark

The original note attempted a benchmark in an infinite-range limit, comparing a Katanin-corrected flow with a tree-level vertex-corrected flow. The running gap was

$$\Delta_{\Lambda} = H \frac{M_0}{M_{\Lambda}}, \quad (31)$$

and the magnetization flow had the schematic form

$$\partial_\Lambda M_\Lambda = \frac{M_0}{\beta N} \sum_{\mathbf{k}, \omega} \frac{\Gamma_{\Lambda, \text{Kat}}^{+-z} \delta(k - \Lambda)}{\Delta_\Lambda + E_{\mathbf{k}} - i\omega}. \quad (32)$$

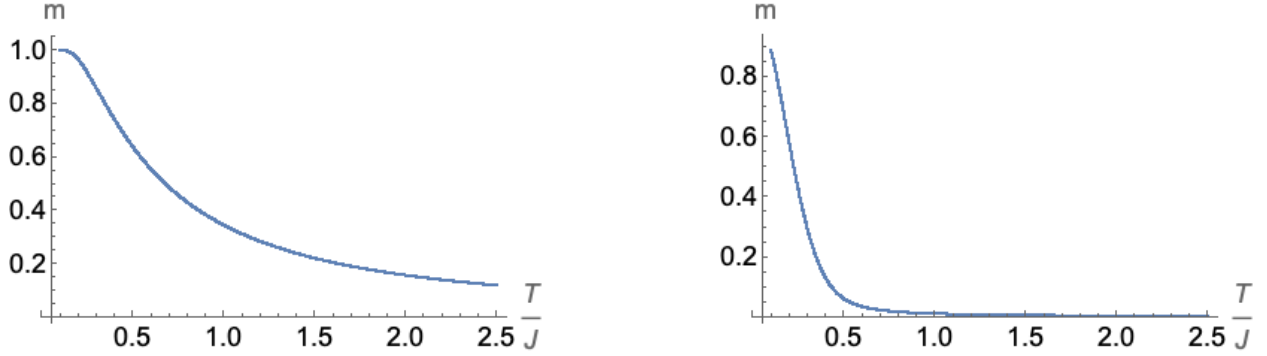


Figure 4: Original diagnostic plots: Katanin correction and vertex correction.

Audit warning. This benchmark is the least polished part of the note. The susceptibility scaling and the $H \rightarrow 0$ limit should be rederived before reuse.

12 Next steps

1. Decide whether the active target is fermionic fRG, spin fRG, or only the Katanin benchmark.
2. Fix the regulator convention and state it once in a boxed definition.
3. Redo the infinite-range benchmark independently in a two-page derivation.
4. Only then decide whether this connects to the quantum-Hall spin-splitting project.

References

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- [3] R. Gersch, C. Honerkamp, D. Rohe, and W. Metzner, “Fermionic renormalization group flow into phases with broken discrete symmetry: charge-density wave mean-field model,” Eur. Phys. J. B **48**, 349 (2005).